Laplace transformu

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt$$

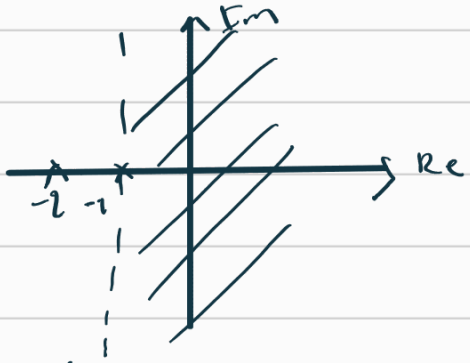
Ex: $X(s) = \dfrac{1}{(s+1)(s+2)}$ — tüm olası ROC'lar = ?

$= \dfrac{A}{(s+1)} + \dfrac{B}{(s+2)} = \dfrac{1}{s+1} - \dfrac{1}{s+2}$

$s=-1 \quad s=-2$

$e^{-at}u(t) \leftrightarrow \frac{1}{s+a}$ $\mathrm{Re}(s) > -a$

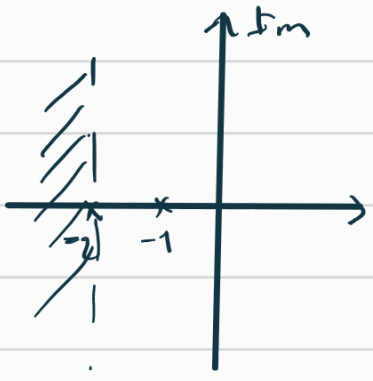
$-e^{-at}u(-t) \leftrightarrow \frac{1}{s+a}$



$x(t)$ sağ yanlı ise

$x(t) = e^{-t}u(t) - e^{-2t}u(t)$

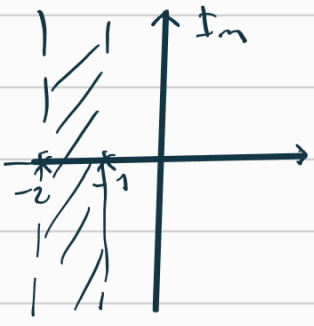
ROC, $s = j\omega$ içerir, FT yakınsar



$x(t)$ sol yanlı ise

$x(t) = -e^{-t}u(-t) + e^{-2t}u(-t)$

FT yoktur



$x(t)$ çift yanlı ise

$x(t) = -e^{-t}u(-t) - e^{-2t}u(t)$

FT yoktur

Ters Laplace Transformu

$$X(s) = x(\sigma + j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-\sigma t}\, e^{-st}\, dt = \mathcal{FT}\{x(t)e^{-\sigma t}\}$$

$$x(t)\, e^{-\sigma t} = \mathcal{FT}^{-1}\{X(\sigma + j\omega)\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\sigma + j\omega)\, e^{j\omega t}\, d\omega$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\sigma + j\omega)\, e^{(\sigma + j\omega)t}\, d\omega$$

$s = \sigma + j\omega \longrightarrow ds = j\, d\omega$

$$x(t) = \frac{1}{2\pi j}\int_{\sigma - j\infty}^{\sigma + j\infty} X(s)\, e^{st}\, ds$$

$$\rightarrow x(s) = \sum_{i=1}^{m} \frac{A_i}{s + a_i} \longrightarrow x(t) = A_i e^{-a_i t}u(t) : \text{sağ yanlı},$$

$$\hookrightarrow x(t) = -A_i e^{-a_i t}u(-t) : \text{sol yanlı}$$

//

Laplace Transformun Özellikleri

$x(t) \overset{\mathcal{L}}{\longleftrightarrow} X(s)$, ROC: R

1) Lineerlik: $\alpha x_1(t) + \beta x_2(t) \longleftrightarrow \alpha X_1(s) + \beta X_2(s)$, ROC: $R_1 \cap R_2$

2) Zamanda öteleme: $x(t-t_0) \longleftrightarrow e^{-st_0} X(s)$, ROC: R

3) S - domainde öteleme: $e^{s_0 t} x(t) \longleftrightarrow X(s-s_0)$, ROC: R+Re{$s_0$}

$s_0 = j\omega_0 \Rightarrow e^{j\omega_0 t} x(t) \longleftrightarrow X(s-j\omega_0)$, ROC: R

4) Zamanda ölçeklendirme: $x(at) = \frac{1}{|a|} X(s/a)$, ROC: R_1

$a = -1 \Rightarrow x(t) \longleftrightarrow X(-s)$, ROC: $-R$

5) Eşlenik alma: $x^*(t) \longleftrightarrow x^*(s^*)$, ROC: R

6) Türev: $\frac{dx(t)}{dt} \longleftrightarrow s X(s)$, ROC: R'yi içerir

$-t x(t) \longleftrightarrow \frac{dX(s)}{ds}$, ROC: R

7) İntegral: $\int_{-\infty}^{t} x(z)\,dz \longleftrightarrow \frac{1}{s} X(s)$, ROC: $R \cap \{Re\{s\} > 0\}$

8) Konvolüsyon: $x_1(t) * x_2(t) \longleftrightarrow X_1(s) X_2(s)$, ROC: $R_1 \cap R_2$

$x(t) \rightarrow \boxed{h(t)} \rightarrow y(t)$

$y(t) = x(t) * h(t) \longleftrightarrow Y(s) = X(s) H(s)$

$H(s) = \frac{Y(s)}{X(s)}$: Transfer fonksiyon

Ex: $x(t) = t e^{at} u(t)$, $X(s) = ?$

$e^{at} u(t) \longleftrightarrow \frac{1}{s-a}$, $Re\{s\} > a$

$$t e^{-at} u(t) \longleftrightarrow -\frac{d}{ds}\left(\frac{1}{s+a}\right) = \frac{1}{(s+a)^2}, \quad Re\{s\} > -a$$

$$\frac{t^2}{2} e^{-at} u(t) \longleftrightarrow \frac{1}{(s+a)^3}, \quad Re\{s\} > -a$$

$$\vdots$$

$$\star \quad \frac{t^{n-1}}{(n-1)!} e^{-at} u(t) \longleftrightarrow \frac{1}{(s+a)^n}, \quad Re\{s\} > -a$$

Ex: $H(s) = \dfrac{2s^2+5s+5}{(s+1)^2(s+2)}$, $Re\{s\} > -1$, $h(t) = ?$

$$H(s) = \frac{A}{(s+1)^2} + \frac{B}{(s+1)} + \frac{C}{(s+2)} \longrightarrow \begin{matrix} A = 2 \\ B = -1 \\ C = 3 \end{matrix}$$

$\uparrow \mathcal{L}^{-1}$

$$h(t) = 2t e^{-t} u(t) - e^{-t} u(t) + 3e^{-2t} u(t) \quad //$$

ROC ile LTI sistemlerin özelliklerinin analizi

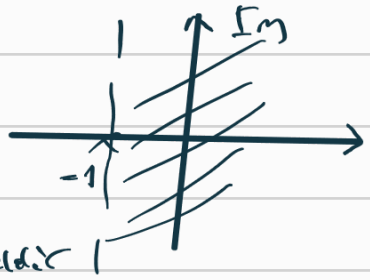
1) **Nedensellik**: $h(t) = 0$, $t < 0$ sağ yanlı $\longrightarrow$ ROC sağ yanlı

2) **kararlılık** : $\int_{-\infty}^{\infty} |h(t)|\,dt < \infty \longrightarrow$ ROC jw eksenini içermelidir.

Ex: $h(t) = e^{-t} u(t)$, kararlı mıdır?

$\updownarrow$

$H(s) = \dfrac{1}{s+1}$, $Re\{s\} > -1$



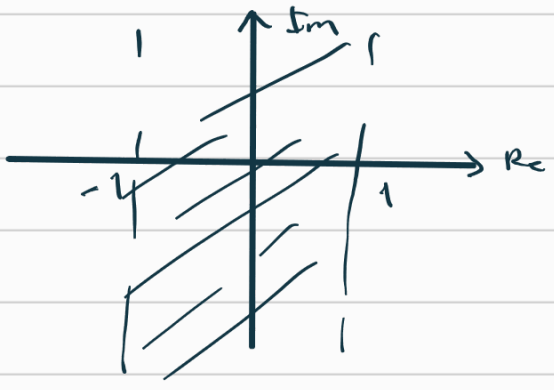
ROC sağ yanlı olduğundan nedenseldir!

jw'yi kesmektedir onun için kararlı //

$x(t) = e^{-b|t|}$

$X(s) = \frac{-2b}{s^2 - b^2}$, $-b < Re\{s\} < b$

Ex: $h(t) = e^{-|t|} \longleftrightarrow H(s) = \frac{-2}{s^2 - 1}$, $-1 < Re\{s\} < 1$

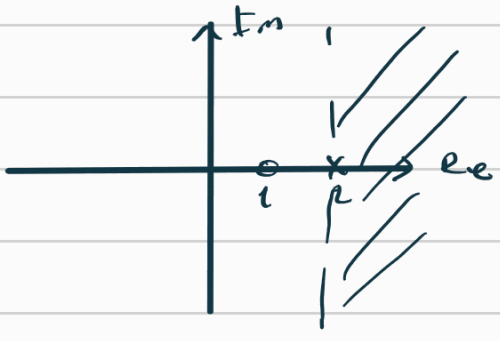


Nedensel değildir

kararlıdır

Ex: $H(s) = \frac{s-1}{(s+1)(s-2)} = \frac{A}{s+1} + \frac{B}{s-2}$, $A = 2/3$

$B = \frac{1}{3}$

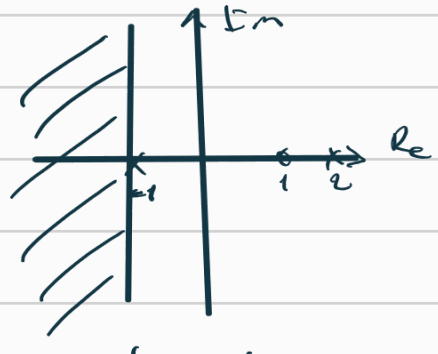


h(t) sağ yanlı

$h(t) = \frac{2}{3} e^{-t} u(t) + \frac{1}{3} e^{2t} u(t)$

Nedensel (sağ yanlı)

kararlı değil

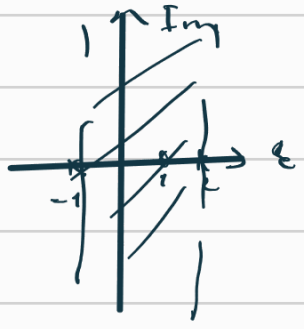


h(t) sol yanlı

$h(t) = -\frac{2}{3} e^{-t} u(-t) - \frac{1}{3} e^{2t} u(-t)$

Nedensel değil Sol yanlı

kararlı değil



h(t) çift yanlı

$h(t) = \frac{2}{3} e^{-t} u(t) - \frac{1}{3} e^{2t} u(t)$

Nedensel değil

kararlı

Ex: $\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt} + 3x(t)$, $h(t) = ?$

$\updownarrow \mathcal{L}$

$s^2Y(s) + 3sY(s) + 2Y(s) = sX(s) + 3X(s)$

$H(s) = \frac{Y(s)}{X(s)} = \frac{s+3}{s^2+3s+2} = \frac{A}{s+1} + \frac{B}{s+2}$, $A = 2$, $B = -1$

$\updownarrow \mathcal{L}^{-1}$

$h(t) = 2e^{-t}u(t) - e^{-t}u(t)$, $Re\{s\} > -1$

z transformu

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$x[n] \overset{z}{\longleftrightarrow} X(z)$

$X(z) = \mathcal{Z}\{x[n]\}$

$\Rightarrow z = re^{j\omega}$ → r: magnitude

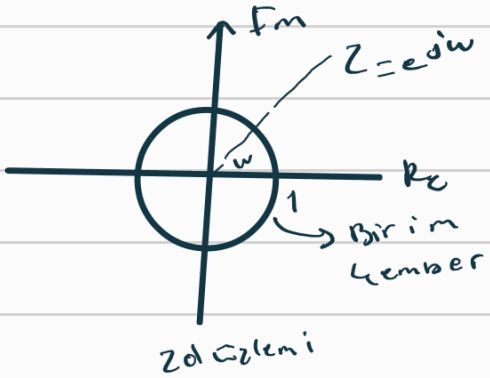
↳ ω: angle

$$X(z) = X(re^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n](re^{j\omega})^{-n} = \sum_{n=-\infty}^{\infty} x[n] r^{-n} e^{j\omega n}$$

$= FF\{x[n] r^{-n}\}$

$\Rightarrow z = e^{j\omega}$ $(r = 1)$

$$X(z) = X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = FF\{x[n]\} = X(z)\Big|_{z=e^{j\omega}}$$



→ Yakınsama bölgesi (region of convergence): z-transform toplamının yakınsadığı z değer aralığıdır.

Not: z-transformun ROC'u birim çemberi içerirse Fourier transformu yakınsar.

Ex: $a^n u[n]$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} = \sum_{n=-\infty}^{\infty} a^n u[n] z^{-n} = \sum_{n=0}^{\infty} (a z^{-1})^n$$

Reminder: $\sum_{k=0}^{\infty} a^k = \frac{1}{1-a}$, $|a|<1$

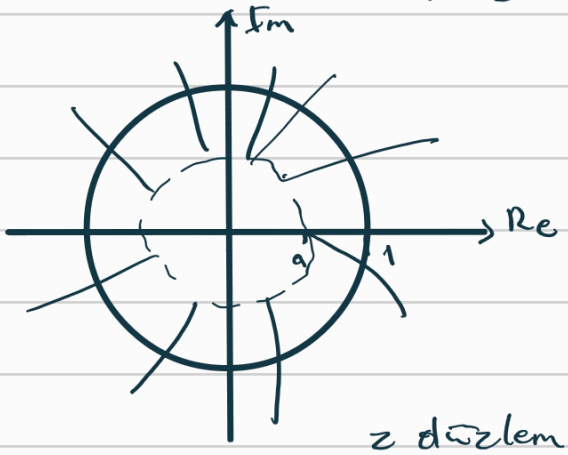
$$X(z) = \frac{1}{1-az^{-1}} = \frac{z}{z-a}, \quad |az^{-1}|<1, \quad |z|>|a|$$

$z=0$ sıfırı

$z=a$ kutupu

$$a^n u[n] \xrightarrow{z} \frac{1}{1-az^{-1}}, \quad |z|>|a| \quad \Big) a=1$$

$$u[n] \longrightarrow \frac{1}{1-z^{-1}}, \quad |z|>1$$



x: kutup

O: sıfır

$0<a<1$

z düzlemi / kutup-sıfır çizimi

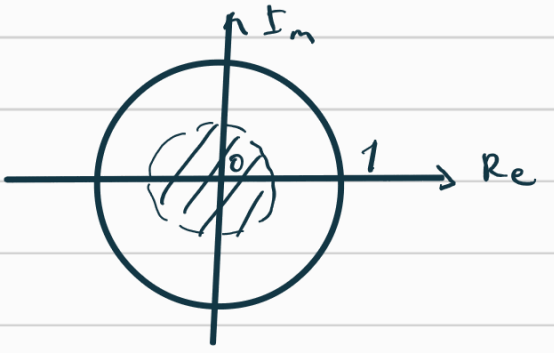
Ex: $x[n] = -a^n u[-n-1]$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} = -\sum_{n=-\infty}^{\infty} a^n u[-n-1] z^{-n}$$

$$= -\sum_{n=-\infty}^{-1} a^n z^{-n} = -\sum_{n=1}^{\infty} a^{-n} z^n = 1 - \sum_{n=0}^{\infty} (a^{-1}z)^n$$

$$= 1 - \frac{1}{1-a^{-1}z} = \frac{1}{1-az^{-1}} = \frac{z}{z-a}, \quad |a^{-1}z|<1, \quad |z|<|a|$$

$-a^n u[-n-1] \xleftrightarrow{Z} \frac{1}{1-az^{-1}}, \quad |z| < |a|$



$0 < a < 1$

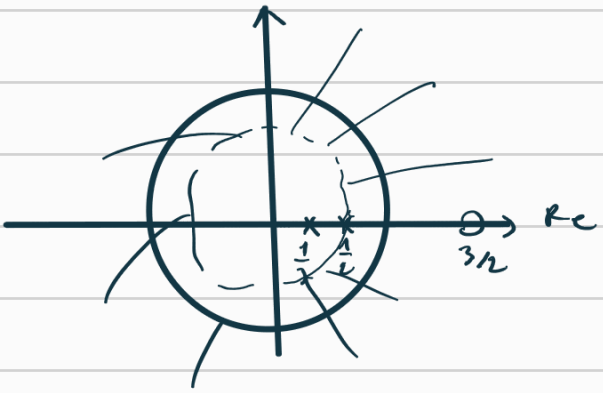
$x(z)$'in ROC'u birim çemberi içermediğinden, $x[n]$'in fourier transformu yoktur.

Ex: $x[n] = 7\left(\frac{1}{3}\right)^n u[n] - 6\left(\frac{1}{2}\right)^n u[n]$

Reminder: $a^n u[n] \xleftrightarrow{Z} \frac{1}{1-az^{-1}}, \quad |z| > |a|$

$$X(z) = \underbrace{\frac{7}{1-\frac{1}{3}z^{-1}}}_{\left|\frac{1}{3}z^{-1}\right| < 1,\ |z| > \frac{1}{3}} - \underbrace{\frac{6}{1-\frac{1}{2}z^{-1}}}_{\left|\frac{1}{2}z^{-1}\right| < 1,\ |z| > 1/2}$$

$$= \frac{1-\frac{3}{2}z^{-1}}{\left(1-\frac{1}{3}z^{-1}\right)\left(1-\frac{1}{2}z^{-1}\right)} = \frac{z(z-3/2)}{(z-1/3)(z-1/2)}, \quad |z| > 1/2$$



Ex: $x[n] = \left(\frac{1}{3}\right)^n \sin\left(\frac{\pi n}{4}\right) u[n]$, $X(z) = ?$

$$x[n] = \frac{1}{2j}\left(\frac{1}{3}e^{j\pi/4}\right)^n u[n] - \frac{1}{2j}\left(\frac{1}{3}e^{-j\pi/4}\right)^n u[n]$$

$$X(z) = \underbrace{\frac{1}{2j}\,\frac{1}{1-\frac{1}{3}e^{j\pi/4}z^{-1}}}_{\left|\frac{1}{3}e^{j\pi/4}z^{-1}\right|<1} - \underbrace{\frac{1}{2j}\,\frac{1}{1-\frac{1}{3}e^{-j\pi/4}z^{-1}}}_{\left|\frac{1}{3}e^{-j\pi/4}z^{-1}\right|<1}$$

$$= \frac{z\,/\,3\sqrt{2}}{\left(z-\frac{1}{3}e^{j\pi/4}\right)\left(z-\frac{1}{3}e^{-j\pi/4}\right)}, \quad |z| > 1/3$$

Not: Payın derecesi > paydanın derecesi → sonsuzda kutup vardır

< → sonsuzda sıfır vardır.

Yakınsaklık bölgesi (ROC) özellikleri:

1) $X(z)$'nin ROC'u z düzleminde

2) ROC, herhangi bir kutup içermez

3) $x[n]$ sonlu süreli ise, ROC, olası $z=0$ ve/veya $z=\infty$ değerleri dışında bütün z-düzlemidir.

Ex: $x[n] = \delta[n]$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} = \sum_{n=-\infty}^{\infty} \delta[n]z^{-n} = 1,$$

$\delta[n] = 1, n=0$

$\delta[n] \xleftrightarrow{Z} 1$, $\forall z$ (ROC, $z=0$ ve $z=\infty$ içeren bütün z düzlemidir)

Ex: $x[n] = \delta[n-1]$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} = \sum_{n=-\infty}^{\infty} \delta[n-1] z^{-n} = z^{-1} = \frac{1}{z}$$

$\delta[n-1] = 1, n=1$

$z=0$' da bir kutup var

ROC, $z=\infty$ içeren, $z=0$ dışında bütün z düzlemidir.

Ex: $x[n] = \delta[n+1]$

$$X(z) = \delta[n+1]$$

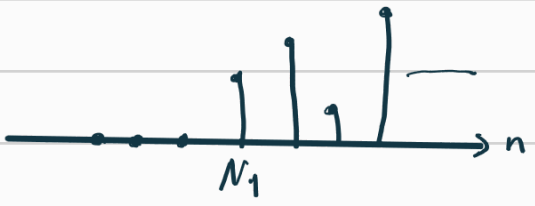
$$X(z) = \sum_{n=-\infty}^{\infty} \delta[n+1] z^{-n} = z$$

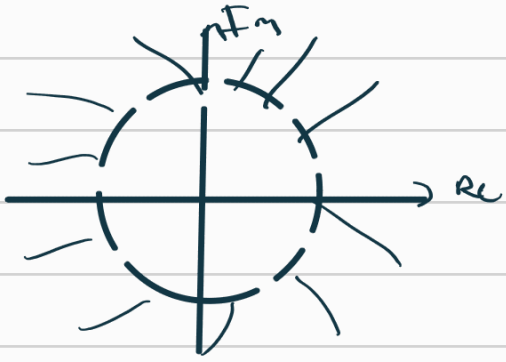
$\delta[n+1] = 1, n=-1$

$z=\infty$' da bir kutup vardır

ROC, $z=0$ içeren, $z=\infty$ dışında bütün z düzlemidir.

4) $x[n]$ sağ yanlı ve $|z|=r_0$ çemberi ROC'da ise, $|z|>r_0$ sağlayan son'lu bütün z değerleri de ROC'dedir.

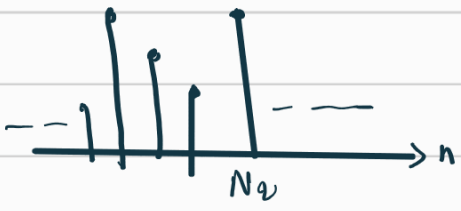


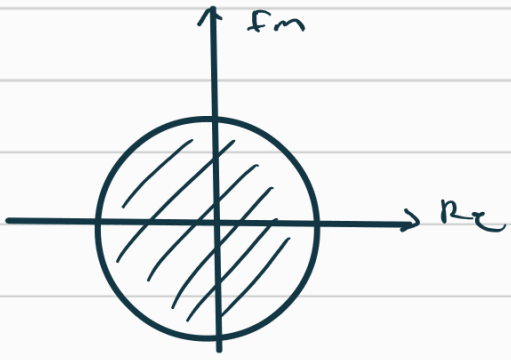


$X(z) = \sum_{n=} x[n] z^{-n}$ → $N_1 < 0$ iken $z \to \infty$, $X(z)$ sonsuzdur

, ve ROC, $z=\infty$ içermez

5) $x[n]$ sol yanlı ve $|z|=r_0$ çemberi ROC'da ise, $0 < |z| < r_0$ sağlayan bütün z değerleri de ROC'dadır olur





6) $x[n]$ iki yanlı ve $|z|=r_0$ çemberi ROC'da ise, ROC, z-düzleminde $|z|=r_0$ çemberini içeren bir halkadır



7) $X(z)$ rasyonel ise, ROC, ya kutuplarla sınırlıdır ya da sonsuza genişler ve kutup içermez

→ $x[n]$ sağ yanlı ise, ROC, en dışındaki kutuplann dışındaki z düzlemidir.

→ $x[n]$ sol yanlı ise, ROC, en içerdeki sıfırdan farklı kutupların içinde z düzlemidir.